$$kv = \omega \; ; \quad \sum_{k} \longrightarrow \frac{L^3}{(2\pi)^3} \int d\vec{k} = \frac{4\pi L^3}{(2\pi)^3} \int_0^\infty k^2 dk = \frac{L^3}{2\pi^2 v^3} \int_0^\infty \omega^2 d\omega$$

$$\underset{\text{per atom}}{\langle E \rangle} = 3\hbar\omega \left(n_B(\beta\hbar\omega) + \tfrac{1}{2} \right)$$

$$\langle E \rangle = 3 \sum_{\vec{k}} \hbar\omega_{(\vec{k})} \left(n_B(\hbar\omega(\vec{k}), \beta) + \tfrac{1}{2} \right)$$

$$= \frac{3L^3}{2\pi^2 v^3} \int_0^\infty \underbrace{(\hbar\omega)\left(n_B(\beta\hbar\omega) + \tfrac{1}{2} \right)}_{f_{(\omega)}} \omega^2 d\omega = \int_0^\infty g_{(\omega)} f_{(\omega)} d\omega$$

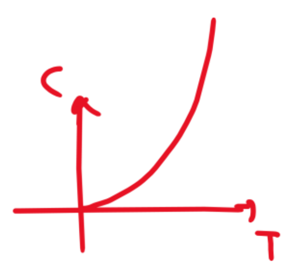
$$\omega^2 \frac{3L^3}{2\pi^2 v^3} = g_{(\omega)}$$

$$g_{(\omega)} = \frac{9L^3\omega^2}{6\pi^2 v^3} = N \frac{9\omega^2}{6\pi^2 n v^3} \qquad 6\pi^2 n v^3 =: \omega_d^3 \qquad n := \frac{N}{L^3} = \frac{N}{V}$$

$$g_{(\omega)} = N \frac{9\omega^2}{\omega_d^3}$$

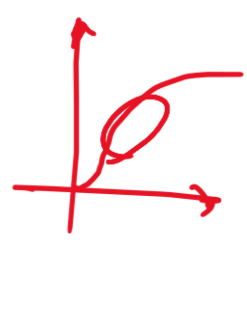
وابستگی به دما ندارد
ولی بی‌نهایت
است.

$$\langle E \rangle = \frac{9N\hbar}{\omega_d^3 (\beta\hbar)^4} \frac{\pi^4}{15} + A \qquad \frac{T^3}{T_D^3}$$



$$C = \frac{\partial \langle E \rangle}{\partial T} = N k_B \frac{12\pi^4}{5} \frac{(k_B T)^3}{(\hbar\omega_d)^3} \propto T^3$$

$$\int_0^{\omega_{cutoff}} g_{(\omega)} d\omega = 3N \qquad \boxed{k_B T_D = \hbar\omega_d}$$



$$\langle E\rangle = \int_0^{\omega_{cutoff}} g_{(\omega)}\, d\omega\, (\hbar\omega)\left(n_B(\beta\hbar\omega) + 1/2\right)$$

$$\langle E\rangle = \int_0^{\omega_{cutoff}} \frac{9N\hbar}{\omega_d^3}\,\omega^3\left(\frac{1}{e^{\beta\hbar\omega}-1} + \underline{1/2}\right) d\omega \;\; ; \;\; \int_0^{\omega_d} \frac{9N\hbar}{2\omega_d^3}\,\omega^3\, d\omega$$

$$\int^{\omega_{cutoff}} g_{(\omega)}\, d\omega = 3N = \int_0^{\omega_{cutoff}} \frac{9N}{\omega_d^3}\,\omega^2\, d\omega = \frac{\overset{3}{\cancel{9}}N}{\omega_d^3}\,\frac{\omega_{cutoff}^3}{\cancel{3}}$$

$$\longrightarrow \boxed{\omega_{cutoff} = \omega_d}$$

حالات حدی :

$$\langle \tilde{E}\rangle = \int_0^{\omega_{cutoff}} g_{(\omega)}\, d\omega\, (\hbar\omega)\left(\underbrace{n_B(\beta\hbar\omega)}\right)$$

$\beta\hbar\omega \ll 1$

$$\frac{1}{\beta\hbar\omega} \leftarrow \frac{1}{e^{\beta\hbar\omega}-1} \simeq \frac{1}{1+\beta\hbar\omega-1}$$

$$\langle \tilde{E}\rangle = \underbrace{\int_0^{\omega_d} g_{(\omega)}\, d\omega}_{3N}\,(k_B T) = 3Nk_BT \Longrightarrow \underline{C = 3Nk_B}$$

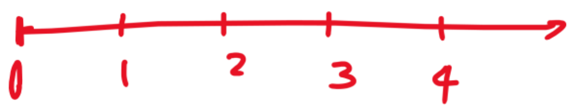
$\beta\hbar\omega \gg 1$

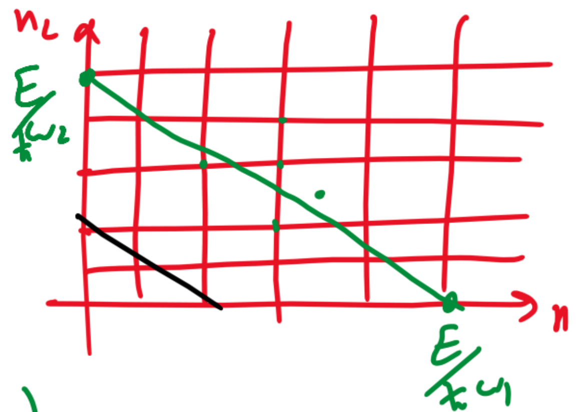
$$\langle E\rangle = \int_0^{\omega_{cutoff}} g_{(\omega)}\, d\omega\, (\hbar\omega)\left(\overbrace{n_B(\beta\hbar\omega)}^{\approx \frac{1}{e^{\beta\hbar\omega}}}\right) + \underbrace{\cancel{\int_{\omega_{cutoff}}^{\infty} g_{(\omega)}\, d\omega\, (\hbar\omega)\left(n_B(\beta\hbar\omega)\right)}}_{0}$$

$$S = k \ln \Omega$$

مدل اینشتین در آنزامبل میکروکانونیک:

$$E_{\vec{n}} = \sum_{i=1}^{3N} \hbar \omega_i n_i \qquad \vec{n} = (n_1, n_2, \dots, n_{3N})$$





$$E = \hbar(\omega_1 n_1 + \omega_2 n_2)$$

$$V \propto r^N$$

$$E = \hbar(\omega_1 n_1 + \omega_2 n_2 + \dots + \omega_{3N} n_{3N})$$

$$A = \frac{E}{\hbar\tilde{\omega}} = n_1 + n_2 + n_3 + \dots + n_{3N} = B$$

$$A = n_1 + n_2 + \dots + n_B \qquad B-1$$

$$\Omega = \binom{A+B-1}{A} = \dots$$

$$S = k \ln \Omega$$

$$\ln N! = N \ln N - N$$

مدل درود : Prude

$t=0$

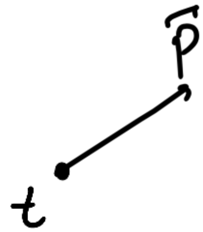
فرضیات نظریه درود:

1) هر الکترون یک زمان پراکندگی (τ) دارد. احتمال پراکندگی $\frac{dt}{\tau}$ است.

2) به محض وقوع یک پراکندگی تکانه الکترون صفر می‌شود. $\vec{p}=\vec{0}$

3) در فاصله بین پراکندگی‌ها الکترون‌ها با بار $-e$ به میدان $\vec{E}$ و $\vec{B}$ خارجی پاسخ می‌دهند.

$$\vec{F} = -e\left(\vec{E} + \vec{v}\times\vec{B}\right)$$



$$\vec{p}_{(t+dt)} = ? = \frac{dt}{\tau}(\vec{0}) + \left(1-\frac{dt}{\tau}\right)\left(\vec{p}_{(t)} + \vec{F}\,dt\right)$$

$$\vec{p}_{(t+dt)} = \vec{p}_{(t)} + \vec{F}\,dt - \frac{\vec{p}_{(t)}\,dt}{\tau}$$

$$\Longrightarrow \boxed{\frac{d\vec{p}}{dt} = \vec{F} - \frac{\vec{p}}{\tau}} \Longrightarrow \frac{d\vec{p}}{dt} = -\frac{\vec{p}}{\tau}$$

$$\vec{p}_{(t)} = \vec{p}_i\, e^{-\frac{t}{\tau}}$$

1) فقط میدان الکتریکی داریم:

$$\frac{d\vec{p}}{dt} = -e\vec{E} - \frac{\vec{p}}{\tau} = 0$$

$$-e\vec{E} = \frac{\vec{p}}{\tau} \Longrightarrow \frac{m\vec{v}}{\tau} = -e\vec{E} \Longrightarrow \boxed{\vec{v} = -\frac{\tau e}{m}\vec{E}}$$

n چگالی الکترون ها در فلز است

$$\vec{j} = -en\vec{v} = \frac{ne^2\tau}{m}\vec{E} = \sigma\vec{E}$$

$$\boxed{\sigma = \frac{ne^2\tau}{m}}$$

$$\vec{v} = \mu\vec{E}$$

$$\boxed{\mu = -\frac{\tau e}{m}}$$

$$\frac{d\vec{P}}{dt} = -e\left[\vec{E} + \vec{v}\times\vec{B}\right] - \frac{\vec{P}}{\tau} = 0$$

$$E = - - -$$

$$\vec{B} = B\hat{z}$$